

Bike Availability & Station Analysis using Power BI

Created By: D. Suganya

Course Name: AI-Driven Data Analytics

Project Overview and Objective

Project Overview

This project focuses on analyzing bike-sharing station data, including bike availability, bike stand capacity, station status, contract-wise performance, and geographical distribution using Microsoft Excel and Power BI. The datasets were cleaned and transformed using Excel and Power Query before being modeled in Power BI. The interactive dashboard provides insights into station utilization, bike availability, operational status, and contract performance to support efficient bike-sharing management.

Objective

The primary objective of this project is to clean, transform, and analyze bike-sharing data using Microsoft Excel and Power BI. The dashboard helps monitor bike availability, evaluate station capacity, analyze operational status, and identify high-performing contracts for better operational planning.

Problem Statement

- Clean and transform raw bike-sharing data by removing duplicates and handling missing values.
 - Analyze bike availability across different contracts.
 - Compare available bikes with total bike stands.
 - Monitor open and closed station performance.
 - Evaluate bonus-enabled stations.
 - Visualize station locations using maps.
 - Develop an interactive Power BI dashboard for operational monitoring and decision-making.
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Tools & Technologies

Power BI

- Power Query
 - DAX Measures
 - Dashboard Development
 - Interactive Visualization
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Data Pre-Processing

Tasks Performed

- Removed duplicate records.

- Handled missing values.
 - Standardized column names.
 - Corrected data types.
 - Removed unnecessary columns.
 - Created calculated columns.
 - Built data relationships.
 - Validated dataset consistency.
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DAX Measures

- Total Stations
 - Total Bike Stands
 - Available Bikes
 - Available Bike Stands
 - Open Stations
 - Closed Stations
 - Bike Availability %
 - Contract Ranking
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Analysis and Visualizations (Power BI)

Dashboard Features

KPI Cards

- Total Stations
- Total Bike Stands
- Available Bikes
- Available Bike Stands
- Open Stations
- Closed Stations
- Bike Availability %

Bar Chart

- Contract-wise Available Bikes

Clustered Column Chart

- Bike Stands vs Available Bikes

Pie Chart

- Open vs Closed Stations

Donut Chart

- Bonus Status Distribution

Table

- Contract-wise Bike Availability Summary

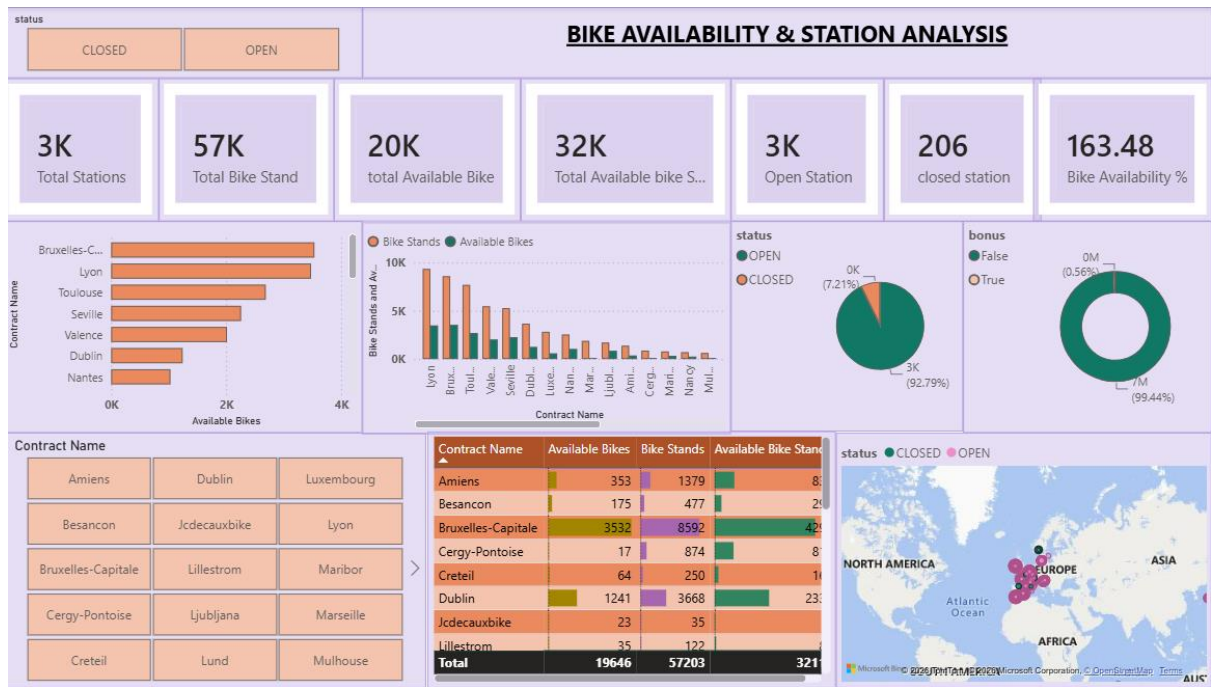
Contract Name	Available Bikes	Bike Stands	Available Bike Stands
Amiens	353	1379	836
Besancon	175	477	299
Bruxelles-Capitale	3532	8592	4297
Cergy-Pontoise	17	874	812
Creteil	64	250	167
Dublin	1241	3668	2334
Jcdecauxbike	23	35	9
Lillestrom	35	122	81
Ljubljana	839	1706	867
Lund	279	517	233
Luxembourg	577	2821	1872
Lyon	3475	9350	5020
Maribor	328	782	452
Marseille	46	1877	658
Mulhouse	13	626	480
Namur	206	437	183
Nancy	241	702	447
Nantes	1029	2538	1470
Rouen	1	555	410
Santander	8	373	327
Seville	2257	5281	2711
Toulouse	2686	7690	4514
Toyama	208	464	254
Valence	2013	5471	3249
Vilnius	0	616	136
Total	19646	57203	32118

Map

- Station Location Analysis

Interactive Features

- Status Slicer
- Contract Name Slicer
- Dynamic Dashboard



Insights & Conclusions

Key Findings

- The dashboard contains approximately 3K bike stations with 57K bike stands available across all contracts.
- A total of 19,646 available bikes were recorded, while 57,203 bike stands were available throughout the network.
- Around 32K bike stands were unoccupied, indicating available parking capacity.
- Open stations account for approximately 92.79% of all stations, while 206 stations (7.21%) are currently closed.
- Bike Availability is 163.48%, indicating strong bike availability across operational stations.
- 99.44% of stations do not provide bonus facilities, whereas only 0.56% support bonus services.
- Bruxelles-Capitale recorded the highest number of available bikes (3,532), followed by Dublin (1,241) and Amiens (353).
- The interactive map highlights station distribution across multiple countries, helping identify operational coverage and regional performance.
- Contract-wise comparison enables stakeholders to identify stations with high bike demand and optimize redistribution strategies.

Recommendations

- Increase bike redistribution in contracts with lower bike availability.

- Monitor closed stations and reduce operational downtime.
- Improve utilization of stations with high bike stand capacity.
- Expand bonus facilities in strategic locations to encourage bike usage.
- Continuously monitor bike availability to improve customer satisfaction and operational efficiency.

Conclusion

The integration of Microsoft Excel and Power BI provided an effective solution for analyzing bike-sharing operations. Excel was used for data cleaning and transformation, while Power BI enabled data modeling, DAX calculations, and interactive dashboard development. The project successfully analyzed 3K stations, 57K bike stands, 19,646 available bikes, 206 closed stations, and 163.48% bike availability, helping stakeholders monitor station performance, optimize bike allocation, and support data-driven operational decisions.